

Athikari Thejesh

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PROFILE SUMMARY

Embedded Firmware Engineer with hands-on experience in bare-metal firmware development, real-time embedded systems, and hardware-software integration using STM32 and ESP32-S3 platforms. Experienced in designing and developing embedded firmware, peripheral driver integration, PCB design, and hardware validation for intelligent connected devices. Proficient in Embedded C, UART, SPI, I2C, I2S, CAN, MQTT, and AWS IoT Core, with a strong foundation in embedded debugging, system integration, and product-oriented firmware development.

TECHNICAL EXPERTISE

Programming : Embedded C, C, Python, 8051 Assembly

Embedded Firmware : Bare-Metal Programming, Firmware Development, Peripheral Driver Development, Interrupt Handling, Context Switching, Task Scheduling, RTOS Fundamentals, DMA Fundamentals, Embedded Debugging

Embedded Platforms : STM32 (ARM Cortex-M3/M4), ESP32, ESP32-S3, ATmega328P, 8051

Communication Protocols : UART, SPI, I2C, I2S, CAN, RS232, MQTT, Wi-Fi

Hardware Development : Sensor Interfacing, Schematic Capture, PCB Layout, Hardware Validation, Circuit Debugging

Development Tools : STM32CubeIDE, Keil µVision, Arduino IDE, KiCad, EasyEDA, Git, MQTT Explorer

EXPERIENCE

Embedded Systems Engineer Intern | SkillArion Development Pvt. Ltd.

June 2026 – Present

- Architected, designed, and developed an ESP32-S3-based Smart Dog Collar from concept to functional prototype, integrating embedded firmware, hardware, and cloud connectivity.
- Developed end-to-end firmware in Embedded C with modular architecture, peripheral drivers, and real-time communication across multiple hardware interfaces.
- Designed the embedded hardware architecture and PCB schematics/layouts using KiCad and EasyEDA, integrating GPS, IMU, digital microphone, audio amplifier, and wireless communication modules.
- Implemented MQTT-based connectivity with AWS IoT Core and performed prototype bring-up, hardware validation, firmware verification, and system-level debugging.

PROJECTS

Electric Vehicle Monitoring & ADAS Warning System | STM32

- Developed a real-time STM32-based embedded system implementing Forward Collision Warning (FCW), Blind Spot Detection (BSD), and Time-To-Collision (TTC) algorithms.
- Designed bare-metal firmware integrating ultrasonic sensors, ADC, UART, PWM, timers, watchdog, and interrupt-driven peripheral drivers.
- Implemented fault monitoring and real-time telemetry for vehicle status acquisition and system health monitoring.
- Implemented interrupt-driven firmware architecture for deterministic real-time event handling and vehicle safety monitoring.

Smart Dog Collar for Real-Time Pet Monitoring | ESP32-S3

- Designed and developed end-to-end Embedded C firmware for an ESP32-S3-based wearable integrating sensor acquisition, audio interfaces, GPS, and wireless connectivity.
- Implemented modular peripheral integration and real-time data handling across I2C, I2S, UART, GPIO, and Wi-Fi interfaces.
- Implemented MQTT-based telemetry and device communication with AWS IoT Core for cloud-connected monitoring.
- Validated integrated hardware through prototype bring-up, peripheral testing, PCB debugging, and system-level firmware verification.

Real-Time Task Scheduler on ARM Cortex-M4 (Bare Metal) :

- Designed and implemented a custom preemptive scheduler using SysTick and PendSV exceptions without external RTOS libraries.
- Developed Task Control Blocks (TCBs), context switching, stack management, and round-robin scheduling for multitasking.
- Implemented interrupt-driven execution using ARM Cortex-M exception handling and NVIC for deterministic task scheduling.
- Performed timing analysis and optimized scheduler latency through low-level firmware debugging.

EDUCATION

B.Tech 🎓 – Electronics & Communication Engineering | Mallareddy Engineering College CGPA: 8.1/10 2023–2027 Hyderabad

CERTIFICATIONS

- Embedded Systems Programming on ARM Cortex-M3/M4 – Udemy
- C Programming – NPTEL ELITE (Silver Medal, 82%)
- Python for Data Science – NPTEL ELITE (Silver Medal, 76%)
- Embedded Systems for Beginners – NIELIT

ACHIEVEMENTS

- First Prize – Arduino Hackathon
- 2nd Runner-Up – DIGILOG Digital Electronics Competition
- First Place – Circuit Innovation Competition